Network Modeling and Analysis in Python

Today's goal • 2/3 items

Module 1
Connectivity in Networks

Module 2
Community Structure in Networks

Module 3
Network Generative Models

The Erdős–Rényi Model

Introduction to Network Generative Models
Video • 7 min

The Erdős–Rényi Model
Video • 15 min

Knowledge Check: The Erdős–Rényi Model
Practice Assignment • Grade: 100%

The Stochastic Block Model

The Stochastic Block Model
Video • 7 min

The Configuration Model

The Configuration Model
Video • 12 min

Knowledge Check: The Stochastic Block Model and The Configuration Model
Practice Assignment • Grade: 100%

Module 3 Wrap-Up

Demo: Generative Models NetworkX Tutorial
Video • 3 min

Module 3 NetworkX Tutorial
Lab • 1h

Module 3 Assignment
Graded Assignment • 30 min

Module 3 Programming Assignment
Programming Assignment • 3h

Optional Readings and Resources
Reading • 10 min

Module 4
Models of Diffusion in Networks

Optional Readings and Resources

0:10 / 0:42 2x

- Book: Network Science by Albert-László Barabási
  - Chapter 3, Random NetworksLink: https://networksciencebook.com/chapter/3
  - Chapter 4, Section 8, Generating Networks with Arbitrary Degree DistributionLink: https://networksciencebook.com/chapter/4#generating-networks
- Newman, M. (2018). Networks. Oxford University Press.
  - Chapter 11: Random GraphsLink: https://doi.org/10.1093/oso/9780198805090.003.0011
  - Chapter 12: The configuration model. Link: https://doi.org/10.1093/oso/9780198805090.003.0012

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